Digital Logic Circuits (Spring 2026)
Quiz #1

1. Determine the decimal values of the signed binary numbers expressed in 2's complement: (a) 01010110; (b) 10101010.
   (a) +86 (b) -86

2. Covert the decimal number 3.248×10^4 to a single-precision floating-point binary number. (Hint: $32480=111111011100000_2$)
   0 10011000 10000100010100110000000

3. Apply DeMorgan's theorems to each expression:
   (a) $\overline{\overline{(\bar{A}+\bar{B})}+\bar{C}}$
   (b) $\overline{(\bar{A}+B)+CD}$
   (a) $(A+B)C$ (b)$A\bar{B}(\bar{C}+\bar{D})$

4. Using Boolean algebra techniques, simplify this expression:
$$AB + A(B + C) + B(B + C)$$
$$B + AC$$

5. Using a Karnaugh map to minimize the following SOP expression:
$$\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D}$$
$$\bar{D} + \bar{B}C$$

6. A positive-going pulse is applied to an inverter. The time interval from the leading edge of the input to the leading edge of the output is 7 ns. This parameter is
   (a) speed-power product (b) propagation delay, t_{PHL}
   (c) propagation delay, t_{PLH} (d) pulse width
   B

7. In a 4-variable Karnaugh map, a 2-variable product term is produced by
   (a) a 2-cell group of 1s (b) an 8-cell group of 1s
   (c) a 4-cell group of 1s (d) a 4-cell group of 0s
   C

8. Develop a logic circuit with four input variables (*A, B, C, D*) that will only produce a 1 output (*X*) when exactly three input variables are 1s.
   X =
$$X = \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$$

9. A VHDL component is called for use in a program by using a
   (a) signal (b) variable
   (c) component instantiation (d) architecture declarations
   C

10. In general, a multiplexer has D
    a) one data input, several data outputs, and selection inputs
    b) one data input, one data output, and one selection input
    c) several data inputs, several data outputs, and selection inputs
    d) several data inputs, one data output, and selection inputs